

The 0.3 keystone: a judgment

the tension dissolves in the x -coordinate

PRIME.RDAGGER.KEYSTONE_JUDGMENT.01 — round 492

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Abstract

Adjudication of the ten-round 0.3 keystone chain (r478–r490), which froze at $\|(1 - P)A^+R_s^{-1}\varphi_1\|/s = 5.86$ against a $5.6 \cdot 10^{-3}$ margin. **U1 IDEA_NAMED.** Four of the five proposed candidates die, three of them structurally: the smooth t -cutoff misdiagnoses the $1/t$ tails (they are forced by $u(\pm L) \neq 0$, i.e. by the compact *time* support, and the enemy symbol $\max\{-\sigma_A, 0\}$ is already Lipschitz, a corner and not a jump); the Fredholm trace-power route cannot start (with the shift, $B = A^- + sI \succeq sI$ makes $K = B^{1/2}R_s^{-1}B^{1/2}$ lie in no Schatten class, since $\lambda_n(R) \asymp \log(n/4L)$, so $(\text{tr } K^m)^{1/m} = \infty$ for every m ; without the shift the traces are not integrals of explicit kernels, because R^{-1} on $L^2[-L, L]$ is the unavailable object); the smoothing preconditioner is incompatible with the support constraint, and the only support-preserving congruence M_w buys ε in the tail against ε^2 in the margin. The named idea is the judge's own: the classical x -space Weil kernel form $Q_W(h) = \int_0^{2L} k(x)(g(0) - g(x)) dx - c_L g(0) + \Pi(h)$ with $k(x) = e^{x/2}/\sinh x$, together with the observation that $I - \text{Re } \tau_x \succeq 0$ for every x , so that *every* truncation of the x -integral is Loewner-safe. There is no off-space term, no tail, no jet, no garbage mode: the remainder is one bounded Hilbert–Schmidt eigenvalue bound with an explicit C^∞ kernel on the compact box $[-L, L]^2$. The form is calibrated against three sealed repo numbers ($\sigma_A(0)$, the r476 Dirichlet hull, the odd sector) and a closing budget is measured: $7.5720 \cdot 10^{-3}$ (even $\lambda_{\min}$) $- 3.34 \cdot 10^{-3}$ (Loewner truncation) $- 3 \times 6.77 \cdot 10^{-4}$ (finite section at $n = 401$) $= 2.2 \cdot 10^{-3} > 0$. **Free diagnosis:** the r488/r490 premise “ R is not coercive, $\inf \text{spec } R = 0$ ” is false; $R \succeq A^+ \succeq 1.568 \cdot 10^{-2} I$ by a two-line Slepian argument. **U2 PROCEED_UNDER_BUDGET** — the keystone is worth it only in the new form, and only under a three-round budget; the frequency form is closed permanently. **U3 REDIRECT_TO_KERNEL_LANE**, with a consolidated six-family anti-list and four new entries. Yoshida–Bombieri zone. No RH claim.

Status. U1 IDEA_NAMED(kernel-Loewner) / U2 PROCEED_UNDER_BUDGET / U3 REDIRECT_TO_KERNEL_LANE.
No promotion of an RH claim.

Claim boundary. Nothing here is $\lambda_*(0.3) \geq c$. This is an adjudication note. It contains no probe, no verifier, no new census and no sealed pin. Every numeral is a float re-measurement made to ground a verdict; interval sealing is gate G1 of the R494 contract. No L^* claim, no $R^\dagger$ claim, no RH claim, no anti-RH claim.

1 Convention

$\text{supp } h \subset [-L, L]$, $g = h * \tilde{h}$, $Q_W = A - P + \Pi$, prime nodes $n \leq e^{2L}$. At $L = 0.3$ one has $2L = 0.6 < \log 2$, so $P \equiv 0$ (scramble-sensitive, vacuous at this L). A is the Fourier multiplier with symbol $\sigma_A(t) = \text{Re } \psi(\frac{1}{4} + it/2) - \log \pi$, $\sigma_A(0) = -5.372183419225666$, $t_c = 6.289835988836903$, normalised by $\langle h, Ah \rangle = (2\pi)^{-1} \int \sigma_A |\hat{h}|^2$. τ_x is translation, $(\tau_x h)(y) = h(y + x)$. The r469 and r479–r490 contracts are in force.

2 U1 — the candidate table

Each candidate is adjudicated on mechanism, hitch, effort and alias against the twenty keystone kills.

Cand.	Verdict	Ground
(a) smooth t -cutoff	KILL, transposed	mechanism misdiagnosed (§2.1)
(b) trace powers / Fredholm	KILL $\times 2$	no Schatten class with the shift; no explicit kernels with it (§2.2)
(c) smoothing preconditioner	KILL	incompatible with $\text{supp} \subset [-L, L]$; ε vs ε^2 (§2.3)
(d) other L / other margin	partial	real but weaker; $L = 0.25$ as calibration rung only (§2.4)
(e) weighted complete basis	accepted inside (f)	such a basis exists and is measured (§2.5)
(f) x -space kernel + Loewner truncation	IDEA _NAMED	§3

2.1 (a) The smooth cutoff

The proposal reads the $1/t$ decay of $\hat{u}$ and the size $|u(\pm L)| \sim 3-18$ as consequences of the sharp band indicator $\mathbf{1}_{|t| < t_c}$. They are not. For any $u \in L^2[-L, L]$ with $u(\pm L) \neq 0$ the transform of the *restriction* decays like $1/t$; this is forced by the compact time support and by nothing else. No symbol smoothing removes it. Moreover the enemy symbol $|\sigma_A| \mathbf{1}_{|t| < t_c} = \max\{-\sigma_A, 0\}$ is continuous with a corner at $\pm t_c$ (a derivative jump, not a value jump), so its kernel already decays like $|x - x'|^{-2}$; an upward mollification over a window w costs only $\approx |\sigma'_A(t_c)| w^2/2 \cdot (2L/2\pi)$ in the enemy trace. Cheap, and irrelevant to a 10^3 deficit. **Kill as stated.** The smoothing is nevertheless the right move — in the *other* variable. Smoothing in x is Loewner-safe and is part of (f); that is where the proposal is redeemed.

2.2 (b) Trace powers

Two independent kills.

(i) *With the shift the operator has no Schatten class.* The r488 test object is $K = B^{1/2} R_s^{-1} B^{1/2}$ with $B = A^- + sI \succeq sI$ and $R_s = A^+ + \Pi + sI$. Since $\sigma_A(t) = \log(t/2\pi) + O(t^{-2})$, the Landau/Weyl count for the compression of $\sigma_+(D)$ to an interval of length $2L$ is $N(\lambda) \approx (2L/2\pi) |\{t : \sigma_A(t) < \lambda\}| = 4Le^\lambda$, i.e. $\lambda_n(R) \asymp \log(n/4L)$. Hence $\lambda_n(K) \asymp s/\log n$ and $\sum_n \lambda_n(K)^m = \infty$ for *every* m : $(\text{tr } K^m)^{1/m} = \infty$. The route cannot start in the shifted formulation.

(ii) *Without the shift the traces are not explicit.* Unshifted, $K = (A^-)^{1/2} R^{-1} (A^-)^{1/2}$ is trace class ($\text{tr } A^- = (2L/2\pi) \int_{|t| < t_c} |\sigma_A| < \infty$ and R is coercive, §4), so the Fredholm inequality $\lambda_{\max} \leq (\text{tr } K^m)^{1/m}$ is available in principle. But $\text{tr } K^m = \text{tr } ((A^- R^{-1})^m)$ contains $R^{-1} = (A^+ + \Pi)^{-1}$ on $L^2[-L, L]$ m times, and that is a finite-interval Wiener–Hopf inverse: precisely the object the whole chain failed to produce. The premise “ $2m$ -fold integrals of explicit kernels, no mode representation needed” is false here. A Loewner minorant $R_0 \preceq R$ with explicit inverse would repair it, but no such R_0 is available for the compression of a multiplier to an interval.

Remark 1. The premise of (b) is nevertheless correct — in the x -coordinate. There the kernels are explicit and elementary. That is a second, independent pointer to (f).

2.3 (c) The smoothing preconditioner

The shift is used through the identity $Q_W = R - A^- = (R + S) - (A^- + S)$, valid for any $S \succeq 0$; a smoothing $S = c(1 + |D|)^\alpha$ therefore lands on the *enemy* as well, and an unbounded enemy destroys the finite-enemy representation, which is the one asset of r486. The alternative is a congruence: $Q_W \succeq 0$ iff $Z^* Q_W Z \succeq 0$ for injective Z with form-dense range. Every Fourier multiplier Z violates $\text{supp} \subset [-L, L]$, which is the defining constraint of the problem. The only support-preserving congruence is multiplication by a weight $w(x)$, and there the scaling is explicit

and adverse: with $w \geq \varepsilon$ at the endpoints the transform tail improves by ε while the margin degrades by ε^2 , so the tail-to-margin ratio degrades like $1/\varepsilon$. **Kill.**

2.4 (d) Another L

Measured in the form of §3 (Legendre Galerkin, $N = 24$; prime-free zone $2L < \log 2$, i.e. $L < 0.346574$):

L	c_L	$\lambda_{\min}^{\text{even}}$	rel. to $ \sigma_A(0) $
0.10	0.90466	$4.571 \cdot 10^{-1}$	$8.5 \cdot 10^{-2}$
0.15	1.35869	$2.181 \cdot 10^{-1}$	$4.1 \cdot 10^{-2}$
0.20	1.69416	$9.493 \cdot 10^{-2}$	$1.8 \cdot 10^{-2}$
0.25	1.96419	$3.336 \cdot 10^{-2}$	$6.2 \cdot 10^{-3}$
0.30	2.19240	$7.573 \cdot 10^{-3}$	$1.4 \cdot 10^{-3}$
0.3465	2.37820	$1.332 \cdot 10^{-3}$	$2.5 \cdot 10^{-4}$

The relative margin really is $4.4\times$ fatter at $L = 0.25$ and $12.5\times$ at $L = 0.20$, and it collapses at the prime-free boundary $(\log 2)/2$ — as it must, λ_* being non-increasing. But a smaller L is a strictly weaker statement, and the declared value of the keystone is the scaling structure. **Partial:** $L = 0.25$ enters R494 as the calibration rung; the target stays $L = 0.3$.

2.5 (e) A basis with summable coupling

Such a basis exists, and the $1/n$ coupling of r481 was an artefact of representing an interval compression of a Fourier multiplier in a basis mismatched to the support — not a property of Q_W . In the x -space form of §3 the plain Legendre Galerkin values at $L = 0.3$ are

N	6	10	14	18	24	36
$\lambda_{\min}^{\text{even}} \cdot 10^3$	7.5968	7.5808	7.5759	7.5740	7.5728	7.5720

— a spread of $2.5 \cdot 10^{-5}$ between $N = 6$ and $N = 36$, monotone from above, no garbage mode anywhere. Compare the frequency lane, where $n = 96$ and $n = 128$ Nyström still carried an $O(1)$ off-space. **Accepted inside (f).**

3 (f) The named idea: x -space kernel, Loewner truncation

3.1 The form

Lemma 1 (x -space form of Q_W for $2L < \log 2$). *For real $h \in L^2[-L, L]$ with $g = h * \tilde{h}$ and $2L < \log 2$,*

$$Q_W(h) = \int_0^{2L} k(x) (g(0) - g(x)) dx - c_L g(0) + \Pi(h),$$

$$k(x) = \frac{e^{x/2}}{\sinh x}, \quad c_L = \int_0^{2L} \left(k(x) - \frac{1}{x} \right) dx + \log(4L) + \gamma + \log \pi,$$

$$\Pi(h) = 2 \left(\langle h, \cosh(x/2) \rangle^2 - \langle h, \sinh(x/2) \rangle^2 \right).$$

Equivalently, since $g(0) - g(x) = \frac{1}{2} \|h - \tau_x h\|_2^2$,

$$Q_W(h) = \frac{1}{2} \int_0^{2L} k(x) \|h - \tau_x h\|_2^2 dx + \Pi(h) - c_L \|h\|_2^2.$$

Derivation. Gauss' integral $\psi(z) = \int_0^\infty (e^{-v}/v - e^{-zv}/(1 - e^{-v}))dv$ at $z = \frac{1}{4} + it/2$, real part, then Plancherel with $(2\pi)^{-1} \int \hat{g}(t) \cos(tv/2) dt = g(v/2)$ and $(2\pi)^{-1} \int \hat{g} = g(0)$, gives $A(h) = \int_0^\infty (g(0)e^{-2x}/x - g(x)k(x))dx - g(0) \log \pi$ after $v = 2x$ and $2e^{-x/2}/(1 - e^{-2x}) = e^{x/2}/\sinh x$. Splitting at $2L$ (where g vanishes) and using $\int_\epsilon^{2L} dx/x - \int_\epsilon^\infty e^{-2x}dx/x \rightarrow \log(4L) + \gamma$ produces c_L . The pole term is $\hat{g}(i/2) + \hat{g}(-i/2) = 2\hat{h}(i/2)\hat{h}(-i/2) = 2(c^2 - s^2)$ with $c = \langle h, \cosh(x/2) \rangle$, $s = \langle h, \sinh(x/2) \rangle$. At $L = 0.3$: $\int_0^{0.6} (k - 1/x) = 0.288137803587$ and $c_L = 2.192404911132$. $\square$

3.2 Calibration against sealed repo numbers

The form is not new mathematics — it is the classical Weil kernel representation. It is new *here*: the ten-round chain worked exclusively in the frequency coordinate. Three independent checks against already-sealed numbers:

Check	this note	sealed
$\sigma_A(0)$ from $\psi(1/4) - \log \pi$	-5.3721834192256654	$[-5.372183419225667, -5.372183419225665]$ (r479)
first Dirichlet $h = \cos(\pi x/2L)$	$1.18025 \cdot 10^{-2}$	hull $[1.150, 1.192] \cdot 10^{-2}$ (r476); midpoint $1.178 \cdot 10^{-2}$ (r486)
odd sector $\lambda_{\min}$	$2.2257 \cdot 10^{-1}$	0.211 (r483), 0.221 (r485), 0.226 (r486)
flat $h \equiv \text{const}$	$1.5836 \cdot 10^{-1}$	—
even $\lambda_{\min}$ (operator)	$7.5720 \cdot 10^{-3}$	—

The Dirichlet hit is the sharp one: an independent representation of the same functional lands inside a hull sealed three rounds earlier. The odd check is weak (the alternative sign convention $\Pi_{\text{odd}} = +2s^2$, as written in r485/r486, gives $2.3787 \cdot 10^{-1}$ — both sit in the sealed range). The sign in Lemma 1 is the one forced by $\hat{g}(\pm i/2) = \hat{h}(i/2)\hat{h}(-i/2)$; pinning that convention is a bookkeeping item of G1, without consequence at this L .

3.3 Why the tension dissolves

Lemma 2 (Loewner monotonicity of the x -truncation). *For every x , $I - \text{Re } \tau_x \succeq 0$ on $L^2[-L, L]$, since $\langle h, (I - \text{Re } \tau_x)h \rangle = \frac{1}{2} \|h - \tau_x h\|^2$. Hence for every measurable $0 \leq w \leq 1$,*

$$Q_W \succeq Q_W^{(w)} := (\kappa_w - c_L) I - T_w + \Pi, \quad \kappa_w = \int_0^{2L} w k,$$

where T_w is the integral operator on $[-L, L]^2$ with kernel $\frac{1}{2} w(|y - y'|) k(|y - y'|)$.

This single line is the whole content of the verdict. The structural tension named in r490 — “low-mode certificates need living tails, completeness needs off-space below the margin” — is a property of the frequency coordinate. In the x -coordinate:

- every truncation has a definite sign, so there is **no residual and no off-space term** to bound, ever;
- the truncated operator is **bounded and Hilbert–Schmidt** once w vanishes near 0, with an **explicit elementary kernel** on the compact box $[-L, L]^2$;
- there is **no infinite frequency range**, no oscillatory integrand, no integration by parts, no endpoint jet, no Slepian, no Nyström nullspace;
- choosing w smooth makes the kernel C^∞ , so the finite-section remainder decays fast and is itself an explicit integral.

3.4 The measured closing budget

Smooth ramp $w = 0$ on $[0, x_0]$, $w = 1$ on $[x_1, 2L]$, quintic C^2 in between. Truncation loss measured against the untruncated $7.5720 \cdot 10^{-3}$:

$[x_0, x_1]$	loss	κ_w	finite margin	$\ T_w\ _{\text{HS}}$
$[10^{-4}, 3 \cdot 10^{-4}]$	$4.8 \cdot 10^{-6}$	—	—	—
$[10^{-3}, 3 \cdot 10^{-3}]$	$8.36 \cdot 10^{-5}$	6.010	$7.489 \cdot 10^{-3}$	—
$[3 \cdot 10^{-3}, 10^{-2}]$	$5.00 \cdot 10^{-4}$	4.8318	$7.0718 \cdot 10^{-3}$	6.378
$[10^{-2}, 3 \cdot 10^{-2}]$	$3.335 \cdot 10^{-3}$	3.6980	$4.2375 \cdot 10^{-3}$	3.519
$[3 \cdot 10^{-2}, 10^{-1}]$	$2.804 \cdot 10^{-2}$	2.5001	$-2.047 \cdot 10^{-2}$	1.748

Hilbert–Schmidt tail of the exact kernel, $(\|T_w\|_{\text{HS}}^2 - \sum_{i,j \leq n} (T_w)_{ij}^2)^{1/2}$ in the orthonormal Legendre basis, with $\|T_w\|_{\text{HS}}^2 = \frac{1}{2} \int_0^{2L} w^2 k^2 (2L - x) dx$ an explicit one-dimensional integral:

$[x_0, x_1]$	$n = 121$	$n = 201$	$n = 301$	$n = 401$
$[3 \cdot 10^{-3}, 10^{-2}]$	$8.17 \cdot 10^{-1}$	$4.79 \cdot 10^{-1}$	$1.08 \cdot 10^{-1}$	$3.87 \cdot 10^{-2}$
$[10^{-2}, 3 \cdot 10^{-2}]$	$4.49 \cdot 10^{-2}$	$8.62 \cdot 10^{-3}$	$1.94 \cdot 10^{-3}$	$6.77 \cdot 10^{-4}$

With the block bound $\lambda_{\max}(S) \leq \lambda_{\max}(P_n S P_n) + 2\|P_n S(1 - P_n)\| + \|(1 - P_n)S(1 - P_n)\|$ and both off-blocks bounded by the tail, the budget at $[x_0, x_1] = [10^{-2}, 3 \cdot 10^{-2}]$, $n = 401$ reads

$$4.2375 \cdot 10^{-3} - 3 \times 6.77 \cdot 10^{-4} = 2.20 \cdot 10^{-3} > 0.$$

A closed, finite certification scheme with a positive margin exists and is measured. It is one 401×401 symmetric matrix whose entries are two-dimensional integrals of an elementary kernel over a compact box, one one-dimensional constant c_L , and one one-dimensional tail integral.

3.5 The honest hitches

1. The quantitative hardness of r490 survives: the required relative precision is $4.2 \cdot 10^{-3}/3.70 \approx 1.1 \cdot 10^{-3}$. The claim is *not* that the problem became easy. The claim is that it became a finite, spectrally convergent, sign-controlled problem instead of a completeness problem.
2. The interval sealing of c_L and of the 401^2 entries is real work (analytic integrand, compact interval, no oscillation — but it must be done, and it is gate G1/G3).
3. The Hilbert–Schmidt tail is a crude bound for an operator norm. If it turns out to be too crude at $n \leq 800$, the lane stops (gate G3 is a hard NO-GO).
4. Lemma 1 is classical, but the program has never written it down; a transcription error would invalidate everything. Hence the three calibration hits are stop conditions, not decorations.

3.6 Alias check against the twenty kills

The route uses no Fourier truncation T , no Slepian, no K_{t_*} , no integration by parts, no endpoint jet, no Nyström nullspace, no shift, no Birman–Schwinger number, no Schur complement against a discrete rest. It touches none of the twenty anti-listed pads. Two new pads are pre-listed in §6 so that they cannot be invented later.

4 Free diagnosis: R is coercive

r488 concludes “ R is positive semi-definite but not coercive” and “the infimum of the form R on $L^2[-L, L]$ is 0”, from $\lambda_{\min}(R) = 0.337, 0.106, 0.076$ at $n = 48, 96, 128$. That inference is false. For $u \in L^2[-L, L]$ and any $t_1 > t_c$, since $\sigma_A \geq 0$ on $[t_c, \infty)$ and is increasing,

$$\langle u, A^+ u \rangle \geq \sigma_A(t_1) \frac{1}{2\pi} \int_{|t| \geq t_1} |\hat{u}|^2 \geq \sigma_A(t_1) (1 - \lambda_0(K_{t_1})) \|u\|^2,$$

because $\lambda_0(K_{t_1}) < 1$ strictly for every finite t_1 . Hence $R \succeq A^+ \succeq c_R I$ with

$$c_R = \sup_{t_1 > t_c} \sigma_A(t_1) (1 - \lambda_0(K_{t_1})) = 1.5685 \cdot 10^{-2} \quad \text{at } t_1 = 8.2.$$

(The Slepian computation is calibrated on the sealed pins: $\lambda_0(K_{t_c}) = 0.8593909212$ against 0.859391, $\lambda_0(K_8) = 0.9350520466$ against 0.935.) Consequences: the shift $s = 0.05$ is not mandatory; the observed n -decay of $\lambda_{\min}(R)$ is convergence to something $\geq 1.57 \cdot 10^{-2}$, not to 0. This does *not* rescue the lane — the residual amplifier becomes $1/c_R = 64$ instead of $1/s = 20$, i.e. worse — but the false premise must not be carried forward as a pin. It is added to the anti-list as a false premise, and it is the reason the unshifted form of (b)(ii) is at least well posed.

5 U2 — is the keystone worth the price?

Yes, but only in the new form and only under a budget. Three facts decide it.

1. **The price collapsed.** In the frequency form the remainder was an ideas problem with no proposal left after ten rounds. In the kernel form the remainder is one bounded eigenvalue bound with a measured closing budget. One to three rounds, not ten.
2. **It dominates the standing fallback.** r471 certified $Q_L \succeq 0$ on the grid family up to $L_{16} = 2.7726$; its named gap F2b is fixed- L density plus a continuity modulus, made necessary because point evaluations are not L^2 -continuous. The kernel lane never evaluates g at a point: it produces $\lambda_*(L) \geq c$ on all of $L^2[-L, L]$ directly, so wherever it executes it makes F2b redundant. F2b is demoted to fallback.
3. **It scales better than the frequency route.** For $2L \geq \log 2$ the prime term enters the same x -space form as an explicit *bounded* perturbation — $\sum_{n \leq e^{2L}} \Lambda(n) n^{-1/2} (\tau_{\log n} + \tau_{-\log n})$ (three terms at $L = 0.8$: $n = 2, 3, 4$), and Lemma 2 is unaffected. Compare the frequency route, where σ_P is almost periodic and the negative set of $\sigma_A - \sigma_P$ is multi-part.

Ranking of internal points by yield over effort after this judgment:

	Point	Effort	Yield
1	R494 kernel lane at $L = 0.3$ (rung 0.25)	1–3 rounds	operator $\lambda_*(0.3) \geq c$
2	same machine up the r471 schedule	2–5 rounds	$\lambda_*(L) \geq c$ at $L = 0.8$ and beyond
3	F2b (r471 density + modulus)	unknown	redundant where 2 executes; fallback
4	F3 Guinand–Weil fixed- L (Lean)	16–28 rounds (r487)	the long pole; unaffected

F3 status, checked as instructed: it is the explicit-formula bridge of the Lean lane. r487 estimated 16–28 rounds (Mellin present in Mathlib, no Guinand–Weil, no ZetaZero API); r489 reduced the density bridge to a single named sorry. Nothing in this judgment touches it; it continues at its own cadence.

The frequency form of the 0.3 keystone — Slepian ladder, Birman–Schwinger, G -certificate, resolvent solve, off-space residual — is closed **permanently**. Ten rounds and twenty pads are enough; there is no eleventh form in that coordinate and none will be accepted.

6 U3 — resource decree

Contract 1 (R494 — PRIME.RDAGGER.KERNEL_LOEWNER.01). *Fixed $L = 0.3$, calibration rung $L = 0.25$. Three gates, in order, A-first.*

G1 (identity, kill gate). *Seal Lemma 1 in interval arithmetic: k , and $c_L \in [2.19240491113, 2.19240491114]$ from $\int_0^{0.6} (k - 1/x) = 0.288137803587$, $\log(4L)$, γ , $\log \pi$. Three mandatory calibration hits, each a stop condition: $\sigma_A(0)$ reproduced from the kernel side to 10^{-14} ; first Dirichlet $1.1803 \cdot 10^{-2}$ inside the sealed r476 hull $[1.150, 1.192] \cdot 10^{-2}$; odd $\lambda_{\min} = 2.226 \cdot 10^{-1}$ against the sealed $0.211/0.221/0.226$, with the Π_{odd} sign convention pinned explicitly. Miss $\Rightarrow$ the identity is mis-transcribed; stop, report, do not proceed.*

G2 (Loewner truncation, budget gate). *Prove Lemma 2 and book the truncation loss in the margin. Working choice $[x_0, x_1] = [10^{-2}, 3 \cdot 10^{-2}]$: $\kappa_w = 3.6980$, loss $3.335 \cdot 10^{-3}$, finite target margin $4.2375 \cdot 10^{-3}$. Forbidden: dropping $\int_0^{x_0}$ without booking the loss. Fail if the loss exceeds one half of the untruncated margin.*

G3 (finite section, load-bearing gate). *Certify $\lambda_{\max}(T_w - \Pi) \leq \kappa_w - c_L - c$ with explicit $c > 0$, via Legendre finite sections and the block bound, with the off-block terms bounded by the exact Hilbert–Schmidt tail $(\frac{1}{2} \int_0^{2L} w^2 k^2 (2L - x) dx - \sum_{i,j \leq n} (T_w)_{ij}^2)^{1/2}$ and the Π tail by the Legendre coefficients of $\cosh(x/2)$, $\sinh(x/2)$. Target: $c \approx 2.2 \cdot 10^{-3}$ at $n = 401$. **GO/NO-GO:** if the certified remainder does not fall below one third of the finite margin at $n \leq 800$, report **STUCK** and close the keystone for good.*

Deliverable on three passes: $\lambda_(0.3) \geq c$ as an operator statement — the first in the program that is neither Galerkin-level nor restricted to a grid subclass. It reproduces the Yoshida–Bombieri zone, so it is an experiments-side note with at most a $[C]$ row; NO RH claim.*

Decree.

1. R494 runs the contract above. Tier: L2 for G1 and G3, L0/L1 for the mechanics. Budget: G1+G2 in one round, G3 in at most two more. Hard stop after R496.
2. The frequency-side 0.3 lane is closed. No round may reopen the A^- Slepian representation, the shifted Birman–Schwinger test, the G -certificate, the resolvent solve, or the off-space residual. The r478–r490 machinery stays sealed as history and may not appear as a pad.
3. The r488/r490 premise “ R not coercive” is marked FALSE PREMISE. No verdict is regressed (they were REDUCED/STUCK, not claims), but the pin does not travel.
4. The Lean bridge lane (r487/r489/r491) is untouched and continues in parallel.
5. The r469 scramble gate stays in force. At $L = 0.3$ the statement is vacuously scramble-insensitive ($P \equiv 0$); from $L \geq (\log 2)/2$ the translations $\tau_{\log n}$ carry the literal $\log n$ and the gate is live.
6. Stop rule: G3 NO-GO closes the keystone for good; resources then go to point 2 of the U2 ranking if G1/G2 passed, otherwise to F2b.

Consolidated anti-list. The twenty keystone kills collapse into six families.

Family	Rounds	Members
F-I majorant pads for the high-band coupling	r478–r485	Fourier–HS majorant; crude κ_{high} ; IBP envelope; Plancherel as $\ Au\ _{[-L,L]}$; flat $c_{\text{tail}}I$ as M_∞ ; co-minorant $\sigma_A(0)K_{t_c} + \Pi$
F-II bases not adapted to the compact support	r481, r484	integer Fourier ONB ($Q_{0n} \sim 1/n$); QR sinc-I Legendre extensions
F-III unresolved discrete modes used as if certified	r483–r485	K_8 jets $\geq \psi_{10}$; Nyström nullspace ONB as the complement; the $n = 48$ leftover $2.76 \cdot 10^{-2}$; the 20 double-step drop
F-IV quadrature and truncation pads	r483–r490	exponential-sum Ci to ∞ ; $T = 400$; GL-tiny at n as the operator tail
F-V wrong Birman–Schwinger objects	r488, r490	$\lambda_{\max}(BG)$ instead of the symmetric product; $3 + \sigma_A(0) \cdot \text{rest}/s = 1.078$ as SATZ; the discriminant certificate as SATZ; $(1 - P_M)A^+u_i/s$ as pad
F-VI false operator premises	r488, r490, r492	unshifted R^{-1} as coercive (correct direction); “inf spec $R = 0$, R not coercive” — refuted, $1.568 \cdot 10^{-2}I$

New entries, pre-listed for the kernel lane:

N1	the x -space Galerkin $\lambda_{\min}$ as if it were the operator value (it is an upper bound)
N2	dropping $\int_0^{x_0}$ without booking the loss in the margin (sign-safe, not margin-safe)
N3	the Hilbert–Schmidt tail used without the $2\times$ off-block term of the block bound
N4	any re-entry of the frequency coordinate (Slepians, Birman–Schwinger, resolvent) into the kernel lane

7 What is not claimed

No $\lambda_*(0.3) \geq c$. This note proves nothing; it adjudicates. Lemma 1 and Lemma 2 are stated as the content of the R494 contract, not as sealed results, and every numeral is a float re-measurement. The r478–r490 verdicts stand unchanged. No L^* claim, no $R^\dagger$ claim, no RH claim, no anti-RH claim.